

# Introduction to musculoskeletal sonography

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# Objectives

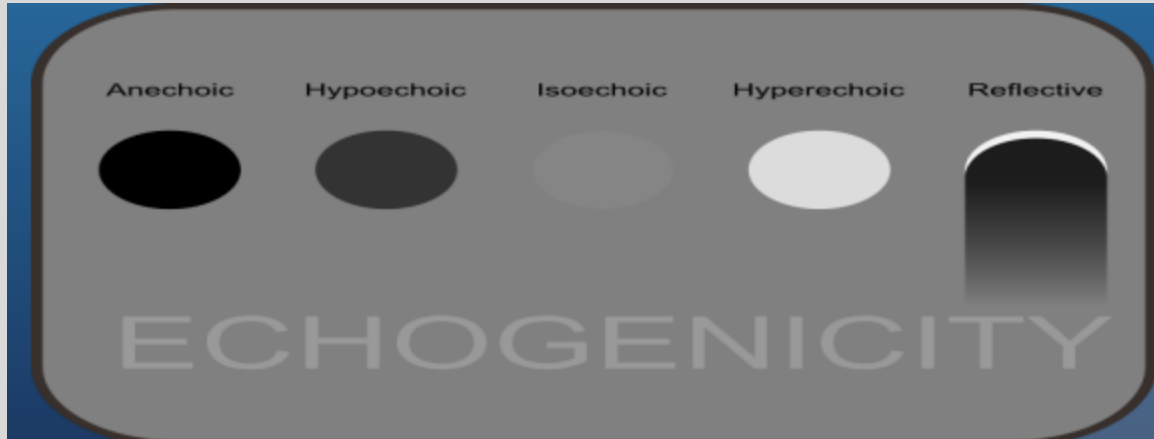
- Student will understand and describe the appearances and differences of tendons, muscles nerves, bone and bursa as well as Anisotropy
- Student will be able to describe common pathologies such as neuropathy and tendinosis
- Student will be able describe indications and limitations of musculoskeletal ultrasound
- Recognize various pathologies presented and compare with normal anatomy

# Sonography advantages

- Readily available and relatively inexpensive
  - often performed as targeted exam
  - can often be preformed to accommodate patient positioning
- Real-time
  - performance of provocative maneuvers, ie. demonstrate tendon subluxation
  - direct visualization during needle placement ☐ Non-ionizing radiation
  - useful in children and during pregnancy
  - Can be used in patients who are not MR compatible, pacemakers or claustrophobia

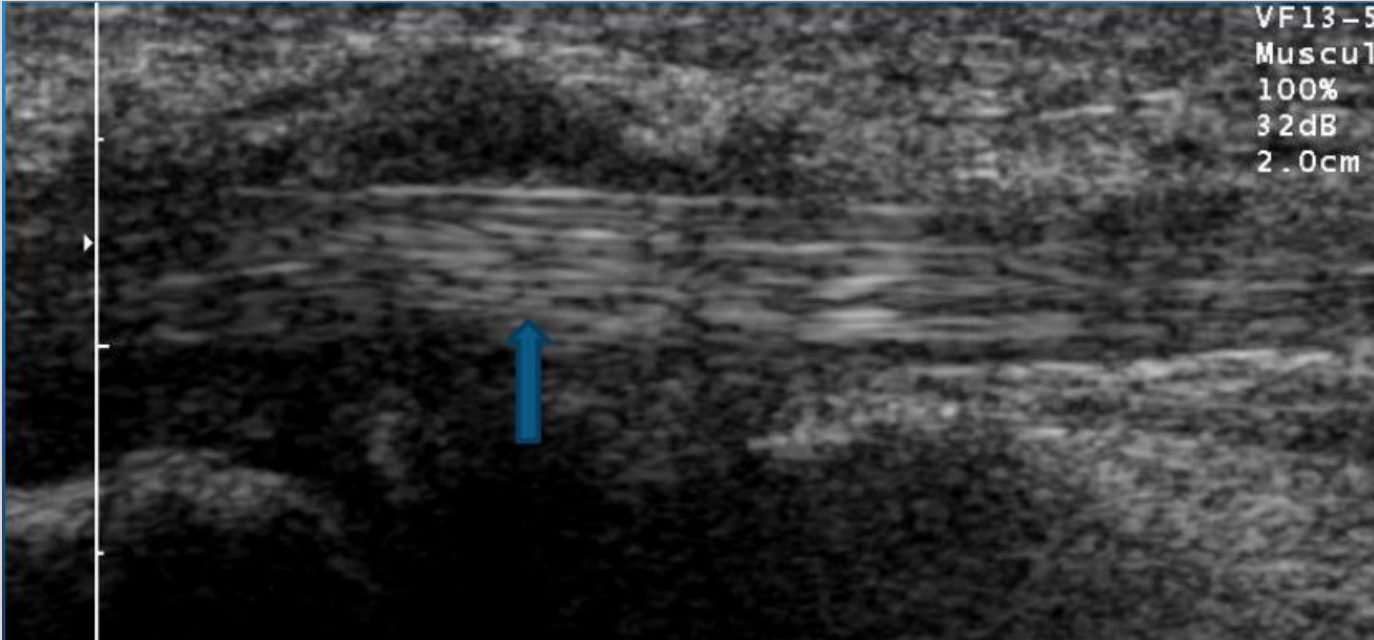
## \*Gray scale terms

- Echogenicity- brightness on the monitor
  - Hyperechoic- increased echogenicity- brighter
  - Isoechoic- similar echogenicity to background tissue
  - Hypoechoic- decreased echogenicity- darker
- Anechoic- black (simple fluid)
- Shadowing (reflective)- the beam is reflected back (calcium/bone)

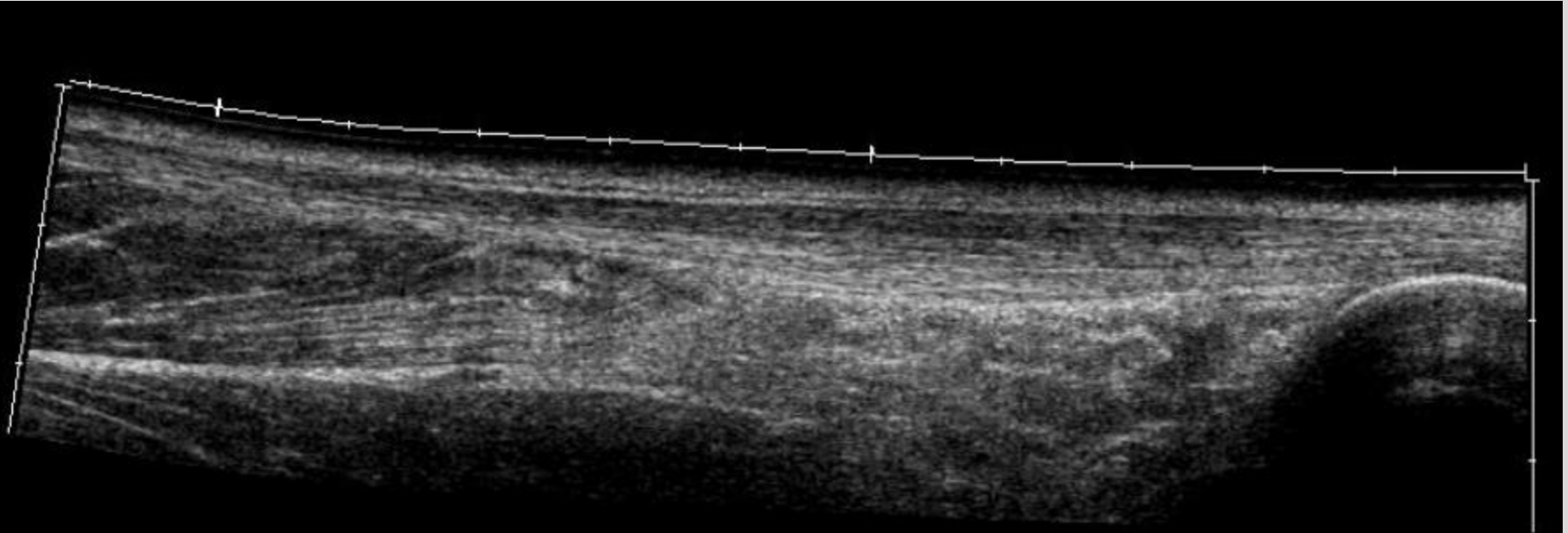


## \*Tendons

- Hyperechoic
- Normally no blood flow noted
- Fibrillar pattern (like a braid of hair) or fiber like



# Achilles Tendon: long axis view



## \*Muscle

- Hypoechoic
- Contains linear hyperechogenic septations fibroadipose septations or perimysium marbled type appearance
- Septations converge to an aponeurosis or tendon on longitudinal plane
- Transverse plane- Starry sky



Arrows- Muscle  
x- tendons

# Ligaments

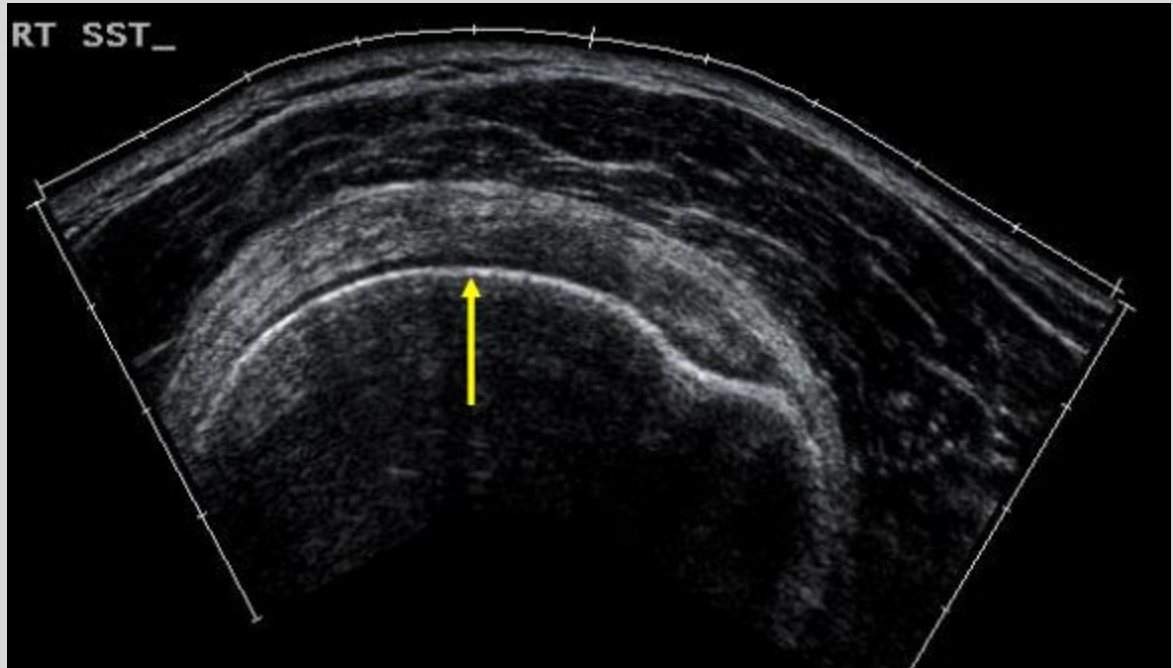
- Hyperechoic appearance, similar to tendons, but more compact than tendons (really dense braided hair)





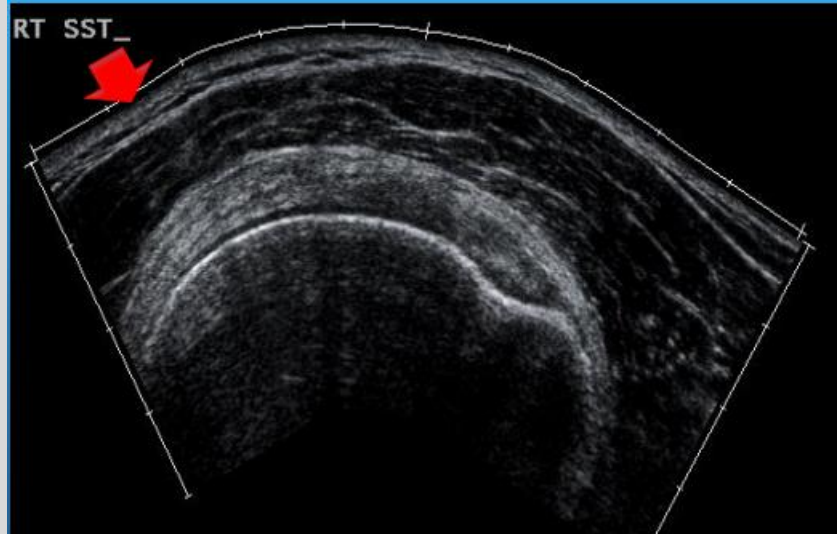
# Bone

- Bright, continuous echo with a smooth contour



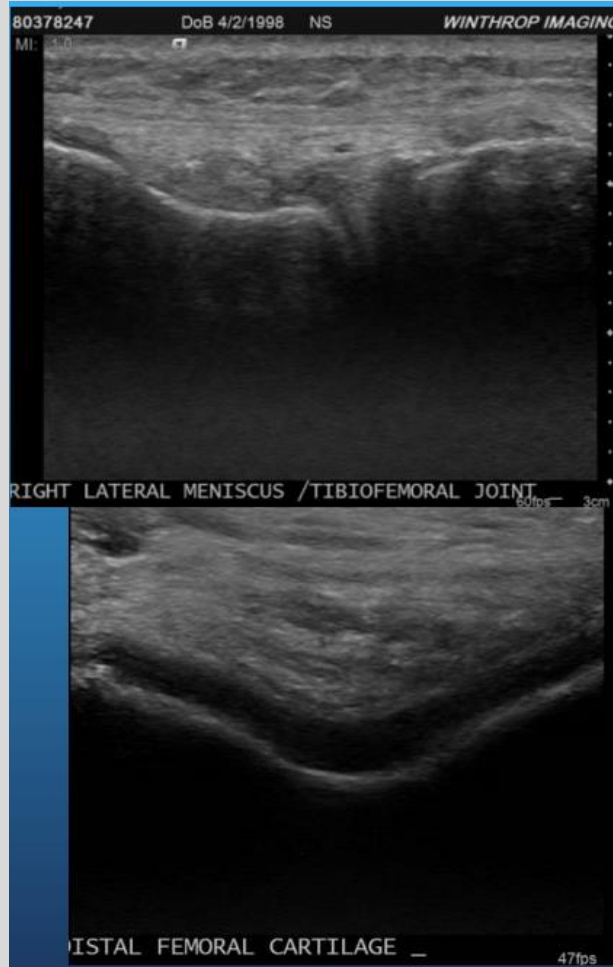
# Fat and skin

- Subcutaneous fat is typically hypoechoic
- Epidermis and dermis- single hyperechoic layer



# Cartilage

- Fibrocartilage is hyperechoic
  - Labrum
  - Menisci
- Hyaline cartilage is hypoechoic
  - ie Patellofemoral cartilage



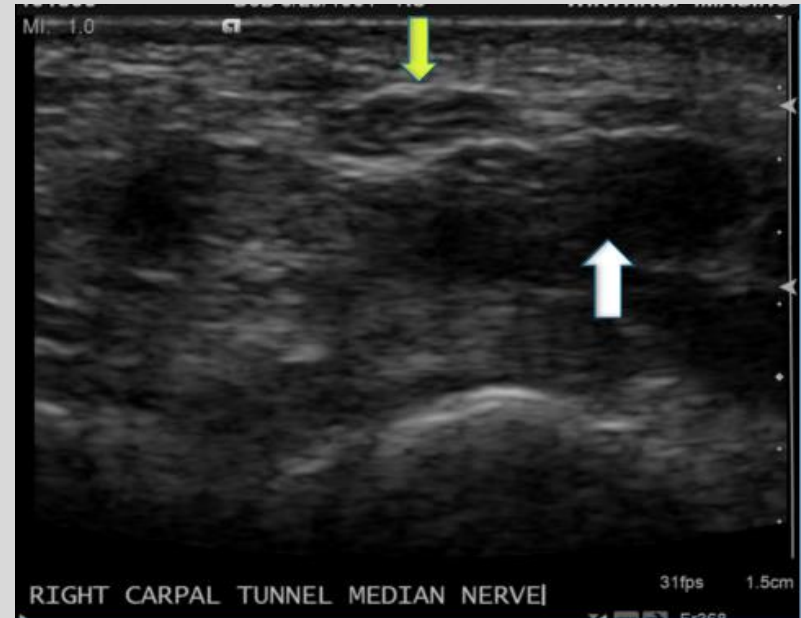
## \*Nerves



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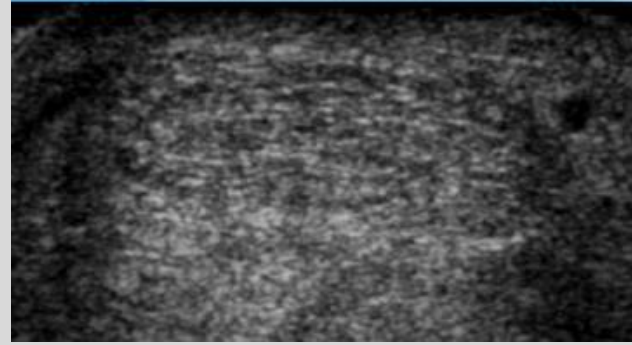
- Round hypoechoic well-defined focus with a homogeneous hyperechoic rim (yellow arrow)
  - Internal hypoechoic foci = nerve fascicles
  - Hyperechoic rim = epineurium.
  - Transverse image- \*Honeycomb image
- Compared to tendons, nerves show differences in echotexture, and tendons tend to be more hyperechoic.
- When in doubt dynamic maneuvers of the fingers or joints are helpful in differentiating between mobile tendons and comparatively static nerves during scanning.
- Do NOT exhibit anisotropy (white arrow tendons



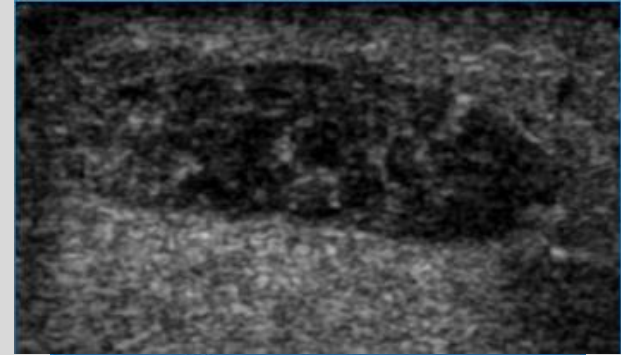
# \*Anisotropy

- Artifact resulting in apparent decreased echogenicity of the tendon or ligament when the probe is NOT 90 degrees to the ultrasound beam • The reason for this is that the probe will only receive the reflected sound if the beam strikes the surface at a right angle
- Can be mistaken for tendinosis
- Common at tendon insertions and when tendons curve

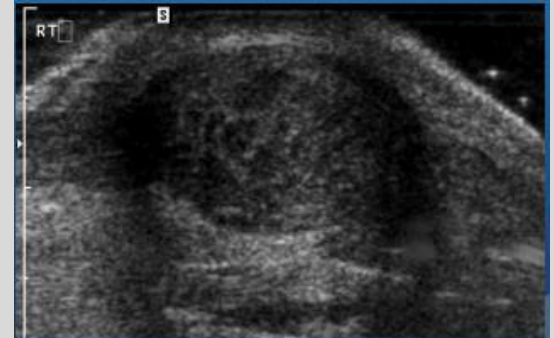
Beam  
perpendicular to  
tendon axis



Beam insonates tendon at  
an angle other than 90  
degrees = Anisotropy



Tendinosis



# Basic pathology

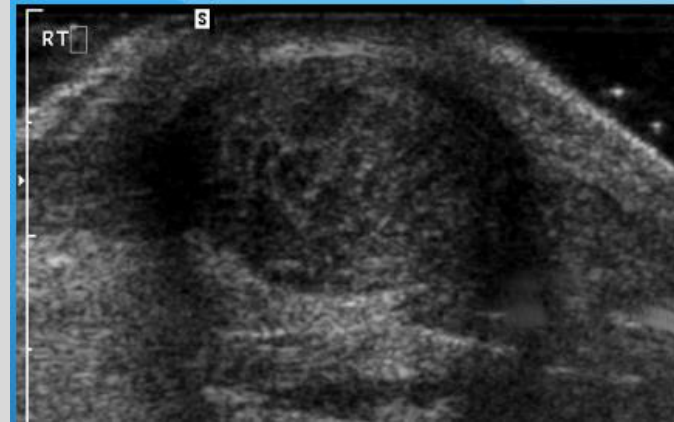


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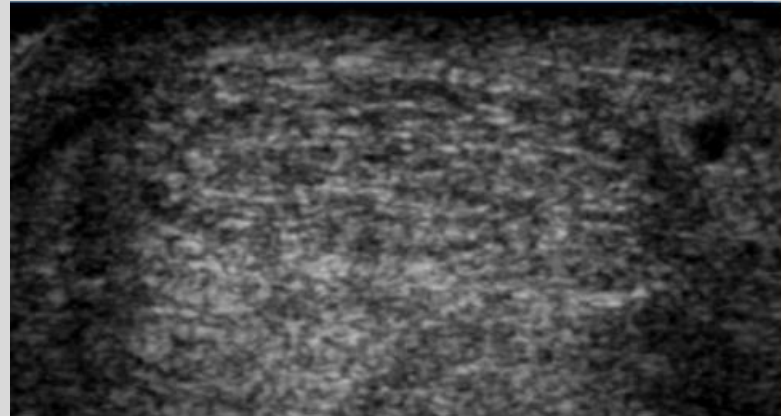
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# \*Tendinosis

- Hypoechoic and typically heterogeneous
- Enlarged or swollen
- Indistinct margins
- May display increased vascularity related to neovascularity



pathic



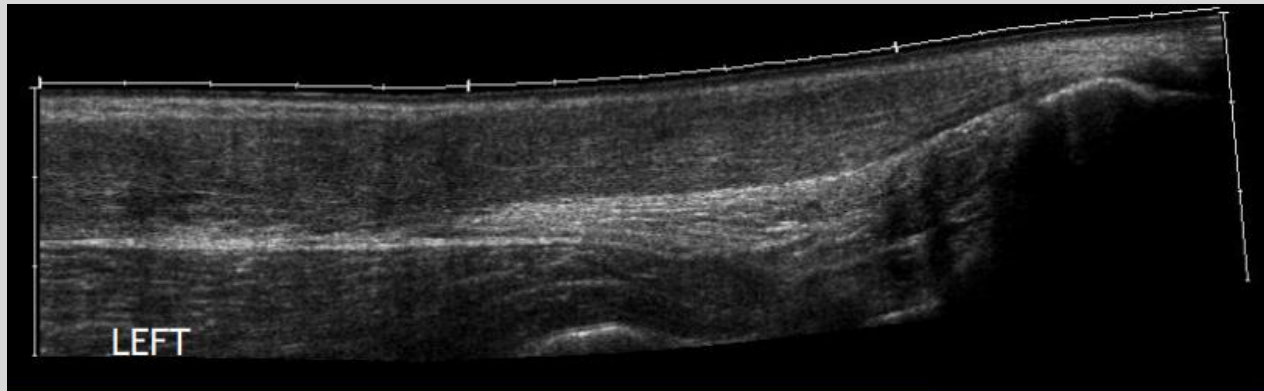
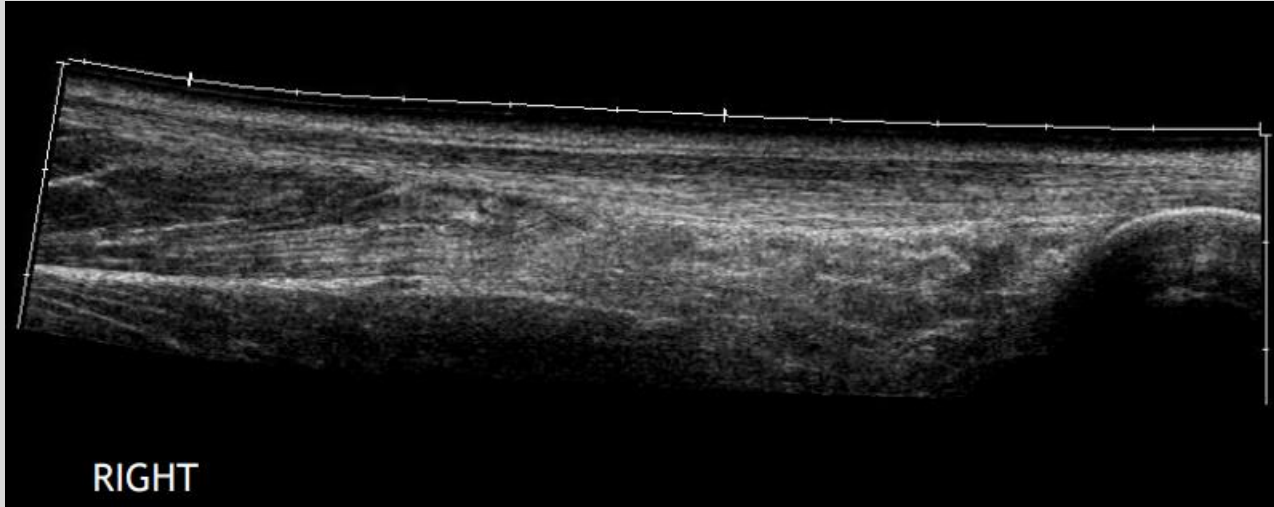
normal



# Achilles Tendon- Which is normal?

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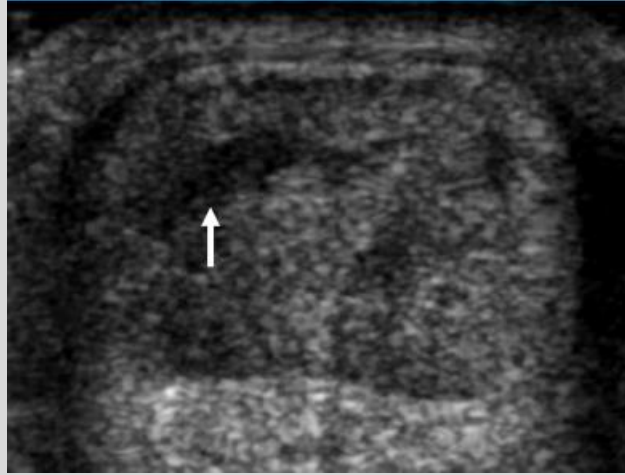
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# Tendon tears

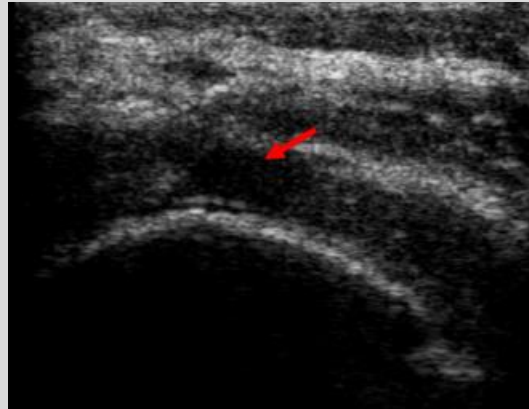
- Partial thickness
  - Intr substance (i.e., longitudinal splits)
  - Does NOT transverse through the entire tendon substance
- Full thickness
  - Extends full thickness of tendon
  - Fluid filled gap



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Transverse view of Achille's tendon demonstrating tendinosis and an interstitial split tear.



Transverse view of supraspinatus tendon demonstrating a full thickness tear anteriorly

# \*Neuropathy

- Sonography entrapments, trauma and tumors
- Dynamic imaging for nerve subluxations ie ulnar nerve
- Neuropathy- increase size and decreased echogenicity when inflamed, often may see a caliber change (Arrow)



# Case 1

- 18 year old female presenting with shoulder pain after attempting cross-fit shoulder press.
- PMH
  - Heart disease with pacemaker
- Limited exam due to pain
- Which imaging test is the next appropriate?
  - X-rays
  - CT
  - MRI
  - Ultrasound

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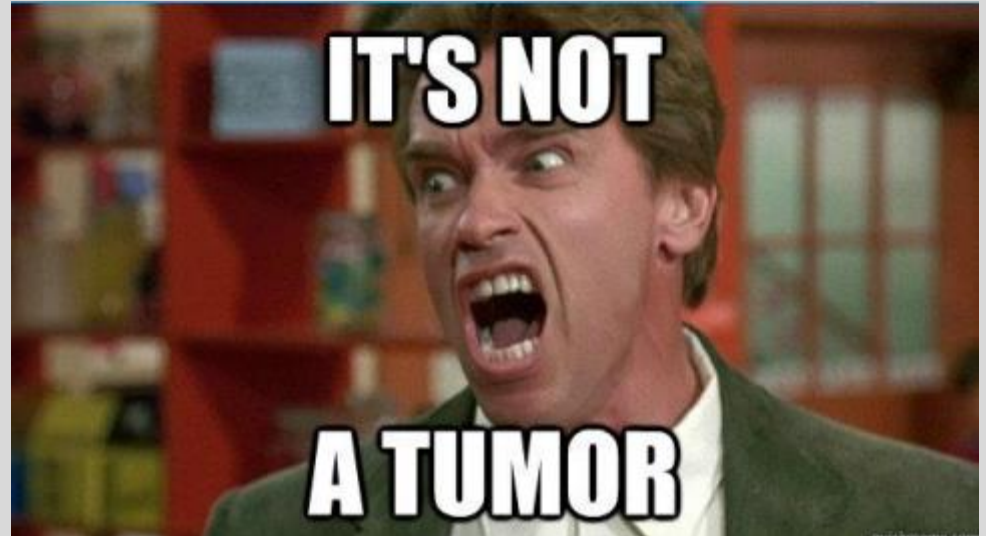


# Case 1

- DDX:
  - Rotator cuff tear
  - Labral tear
  - Fracture?
  - Calcific tendinitis
  - Tumor
    - ... its not a tumor

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# Case 1

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# What did we rule out?

- Greater tuberosity fracture
- Calcific tendinosis
- Tumor (well somewhat)

# What next?

- CT-
  - Radiation
  - Good for fractures
  - CT arthrogram for rotator cuff and labral tears
  - Routine CT NOT useful for imaging the rotator cuff or labrum
- MRI
  - Gold standard for rotator cuff and labral pathology
  - NOT in this case... why?
- Ultrasound
  - Ideal in this patient
  - No radiation
  - Dynamic testing
  - NOT good for osseous pathology
    - That's well not entirely true

# What can/should we look at with ultrasound?

- Rotator cuff
  - What are they?
- Biceps tendon
- AC joint
- Glenohumeral joint
- Nerves



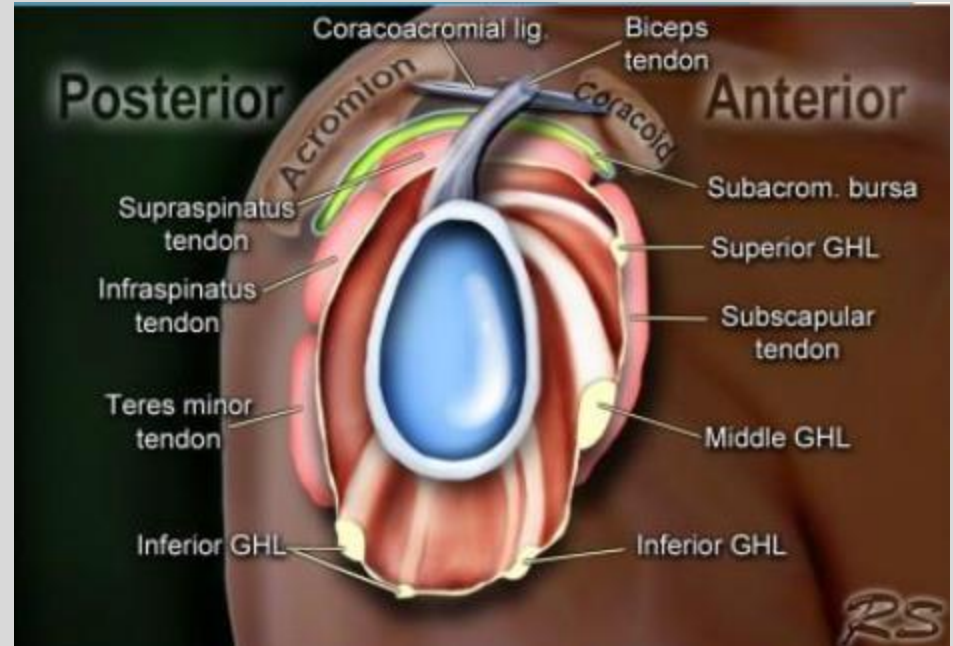
# Rotator Cuff Muscles “SITS”

- Supraspinatus: within suprascapular fossa and inserts upon the greater tuberosity of the humerus.
- Infraspinatus: posterior to scapula inferior to the scapula spine and inserts onto the posterior aspect of the greater tuberosity.
- Teres Minor: inferior to infraspinatus and also inserts onto the posterior aspect of the greater tuberosity.
- Subscapularis: anterior to the scapula inserting onto the lesser tuberosity.

# Shoulder Anatomy: Lateral/Sagittal MRI Correlation

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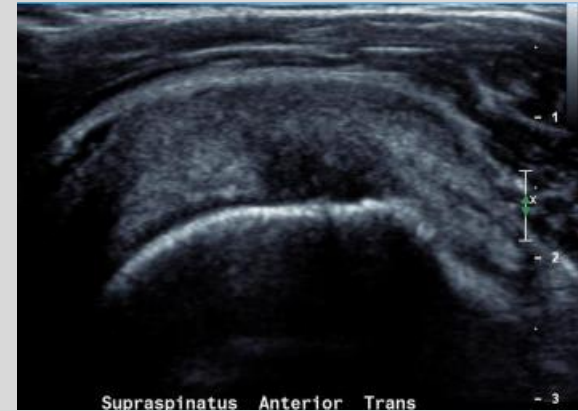
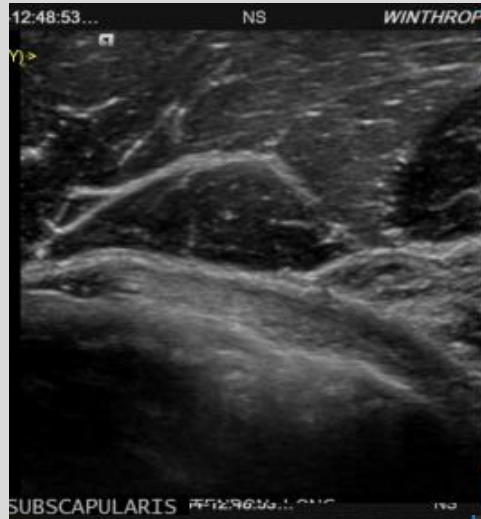
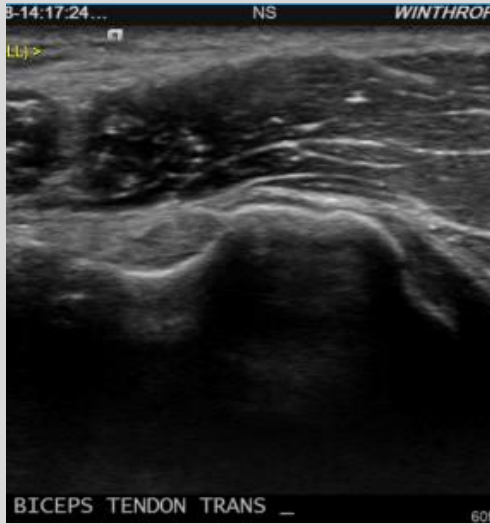
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# US Examination: right shoulder

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Hmm...what can we do next to make sure?

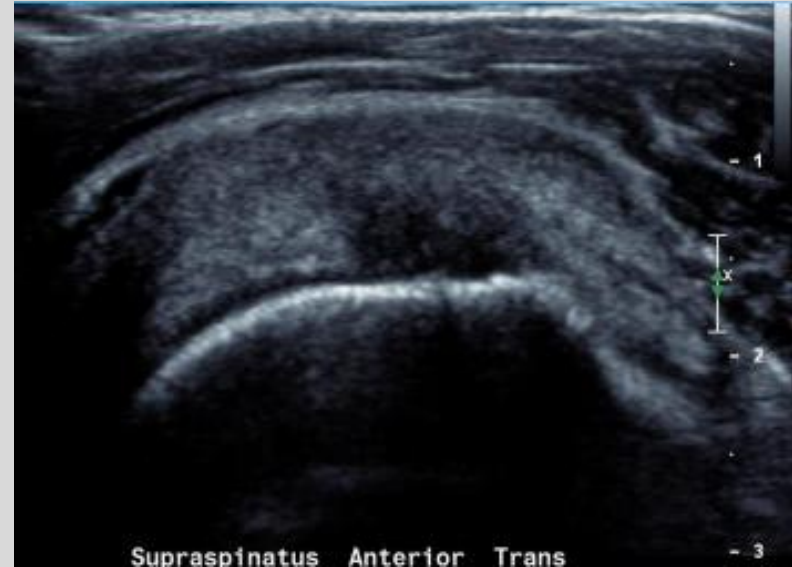
# US Examination: right shoulder

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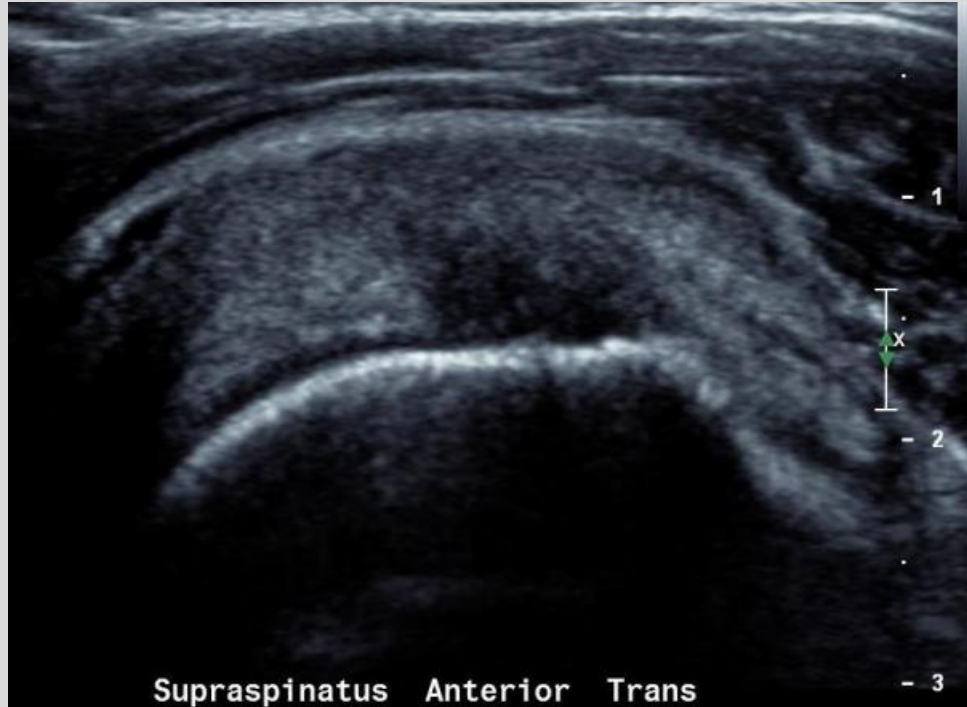


Left



Right

# High grade partial tear



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# Case 2

- 40-year-old rheumatologist who injured their hand tickling their child.

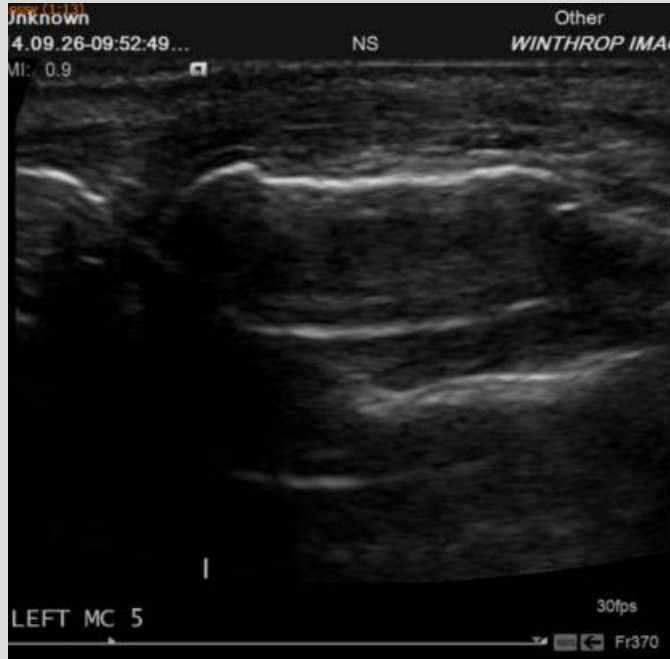


- Ddx
  - Collateral ligament injury
  - Flexor or extensor tendon injury
  - Fracture
  - Pulley injury

# Case 2

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# Case 2

Hmm



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LEFT



H: 0.30  
F: 0.50

STE: 12

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# Dx fracture

Cortical discontinuity

On ultrasound and x-ray trace the cortices around and they should be continuous and smooth

# Case 3

- 45 year old female with increased pain and swelling behind the right knee
- No PMHx
- Refuses to take x-rays because of radiation

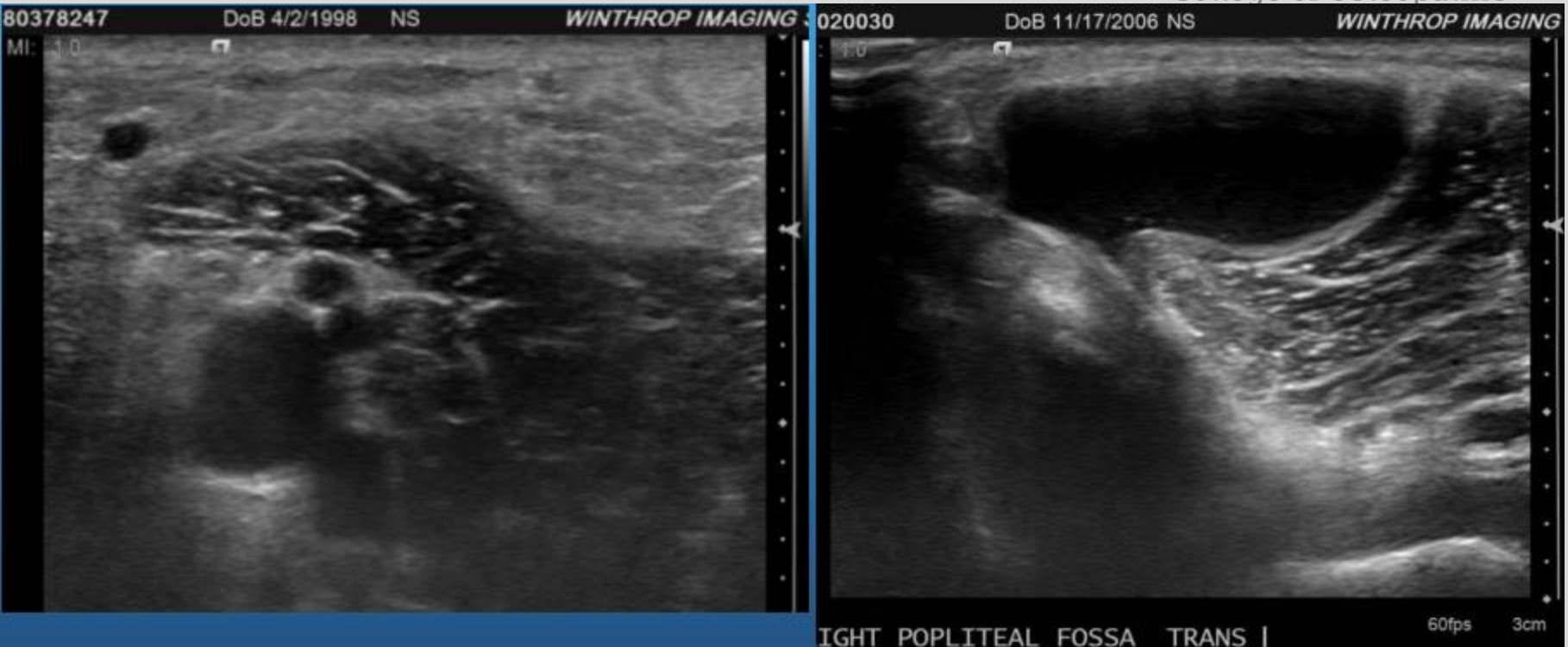


- Ddx
  - \*\* DVT
  - Mass
  - Malignant
  - Sacroma
  - Lipoma
  - Nerve sheath tumor
  - Cyst

# Posterior Knee/Popliteal Fossa Anatomy

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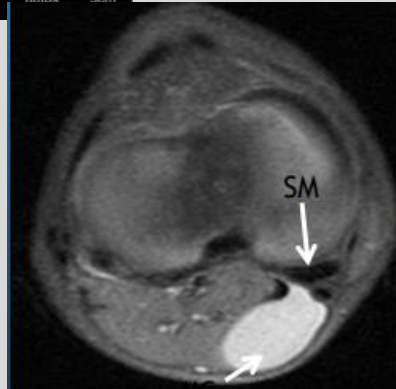


Left

Right

# Popliteal “Bakers” Cyst

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Bursa/neck of “cyst” between the semimembranosus tendon medially and medial head gastrocnemius laterally adjacent to posterior aspect of the medial femoral condyle.

# What can be done for this?



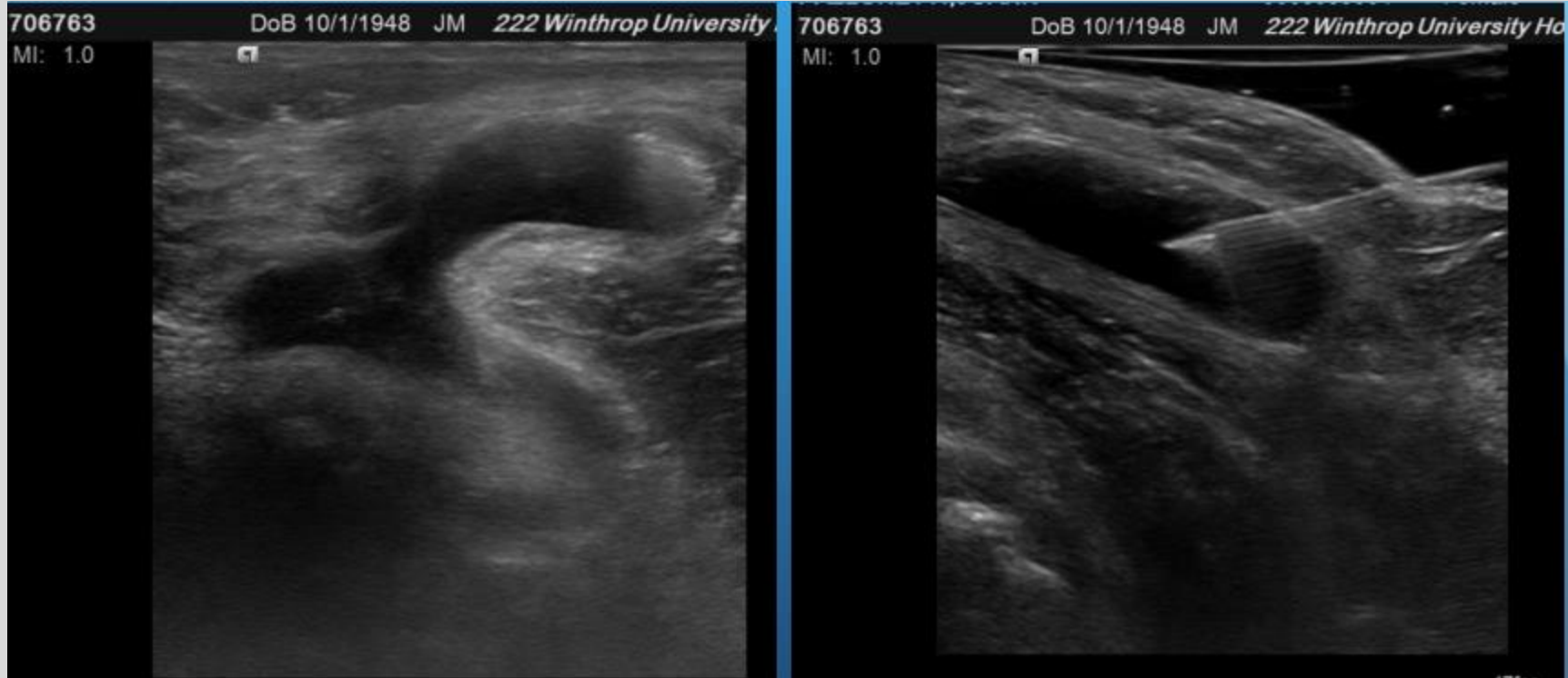
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# Popliteal Cyst

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# Case 4

- 35 year old weekend warrior with posterior ankle/calf pain and swelling.



## Ddx

- DVT\*\*
- Achilles tendon pathology
- Plantaris tendon tear
- Muscle strain/herniation
- Plantar fascia



# Case 4



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# Case 4

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Right symptomatic side



Left asymptomatic side



# Dx Achilles tendinopathy without tear

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# Conclusions

- MSK US – dynamic exam with history taking ☐  
Should be performed by the radiologist or by well trained technologist
- Anatomic location, relationship to adjacent structures, and history are usually more helpful in rendering a diagnosis, rather than standard imaging features

# Lecture and Lab Feedback Form:

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